FIITJEE Students From Classroom / Integrated School Programs have secured to 2 Zonal, 6 State & 18 City Topper Ranks. 33 in Top 100, 78 in Top 200 and 205 in Top 500 All India Ranks bagged by FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2016

ANSWERS, HINTS & SOLUTIONS

FULL TEST–II
(Paper-2)

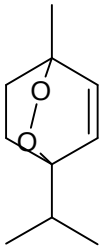
Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A	A
2.	C	C	D
3.	B	B	A
4.	C	B	B
5.	B	C	D
6.	D	A, B, C, D	B
7.	C, D	A, B, C, D	A, B, C
8.	C, D	A, D	B, C
9.	A, C, D	A, B, C, D	B
10.	D	B	C
11.	A	B	B
12.	A	B	B
13.	C	A	C
14.	D	A	B
1.	(A) $\rightarrow$ r, (B) $\rightarrow$ p, q, (C) $\rightarrow$ r, s, (D) $\rightarrow$ p, q	(A) $\rightarrow$ r, (B) $\rightarrow$ p, q, t, (C) $\rightarrow$ r, s, t, (D) $\rightarrow$ r, s, t	(A) $\rightarrow$ s, (B) $\rightarrow$ q, (C) $\rightarrow$ p, (D) $\rightarrow$ r
2.	(A) $\rightarrow$ p, s, (B) $\rightarrow$ q, (C) $\rightarrow$ p, r, (D) $\rightarrow$ q	(A) $\rightarrow$ p, t, (B) $\rightarrow$ s, t, (C) $\rightarrow$ p, q, s, t, (D) $\rightarrow$ p, q, t	(A) $\rightarrow$ p , (B) $\rightarrow$ p, (C) $\rightarrow$ s, (D) $\rightarrow$ q, r
1.	3	5	6
2.	0	0	0
3.	1	5	1
4.	1	9	3
5.	3	6	9

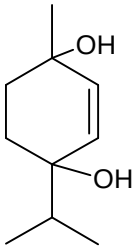
Chemistry

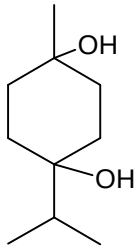
PART – II

SECTION – A

1. $Fe(OH)_3$ is not soluble in excess NaOH.
2. C
$K_2SO_4.Cr_2(SO_4)_3.24H_2O$
3. B
$C=\frac{P}{RT}=\frac{n}{V}$
4. B
$\frac{hC}{\lambda}=-RhC\left[\frac{1}{n^2}-1\right]$
5. C
Here attraction is zero , so potential energy can't be less than 0.
7. A,B,C,D
Ni^{2+} complex becomes $3d^8$ with all e^- paired.
8. A, D
Chiral molecules do not have plane of symmetry
10. B
+vely charged colloid and –vely charged solution
11. B
Neutralisation of charge of colloidal particle.
12.

A= 

B= 

C= 

Section –C

1.
$\frac{31.82-31.78}{31.82};\frac{3\times\text{mole of solute}}{\text{moles of solvent}}$
$\frac{\text{mole of solute}}{\text{moles of solvent}}=\frac{0.00125}{3}$
$[MCl_2]=0.0233$
$K_{SP}=5\times10^{-5}$
2. In excited state it is $\pi_{2P_X}{}^2$
3. Total 15 geometrical isomers.
4. 9
5. A=CO, B=HCN, Y=$CaCN_2$

Mathematics

PART – III

SECTION – A

1. Use optimization technique

2. $y = mx + \frac{2}{m}$ to satisfy in $xy = 1$
   Put D = 0 to get tangents

3. $f(t) = t(t+1)(t+2)(t+3)$ where $t = x^3$
   $f(t) = (t^2+3t)(t^2+3t)+2$
   Use maxima minima concept

6. Use $\overline{OB} = \lambda(\overline{OA}) \times (\overline{OA} \times \hat{j})$

9. Use graph

10. Use binomial triad for probability

(12 – 14)
Use rotation of complex numbers.

Section –C

2. Replace x by (1-x) solve to get the result

3. See minimum number of unknowns